



Department of Electrical and Electronic Engineering

Name of the Experiment: Verification of Thevenin's Theorem

Submitted by

Name: Abdullah Al Junaid

ID: 2024-3-60-082

Section: 10

Department: Computer Science and Engineering

Submitted to

Dr. Saeed Mahmud Ullah

Adjunct Professor, Department of Electrical and Electronic Engineering

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Saeed Mahmud Ullah
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Theory:

Thevenin's Theorem states that any complex linear DC network can be replaced by an equivalent circuit consisting of a single Thevenin voltage source (V_{th}) in series with a single Thevenin resistance (R_{th}).

The load current (I_L) for both the original and equivalent circuits is defined by:

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

Equipments:

1. Variable DC power supply
2. Voltmeter and ammeter
3. Resistances
4. Breadboard
5. Connecting wires.

Circuit Diagram:

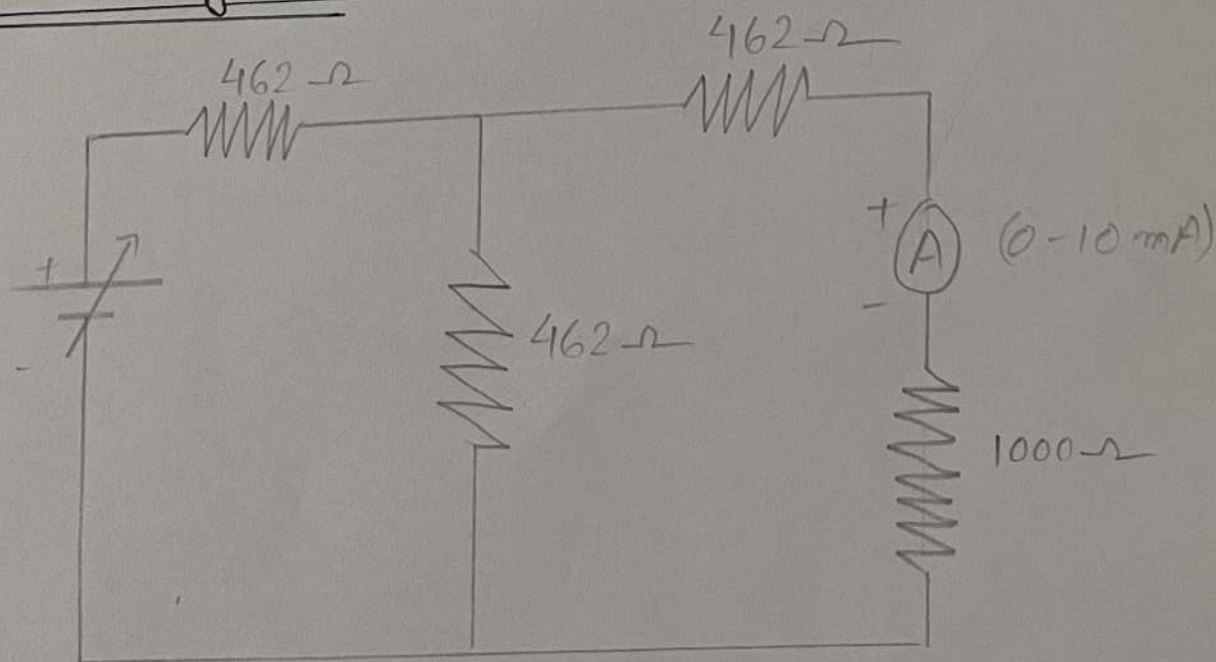


Figure: To Find load current

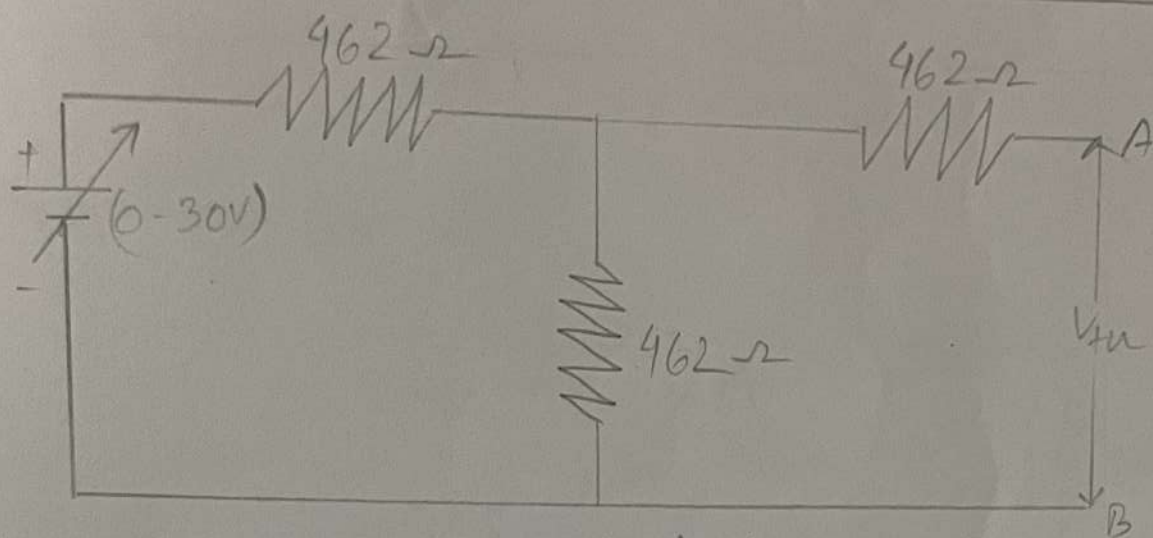


Figure: To find V_{th}

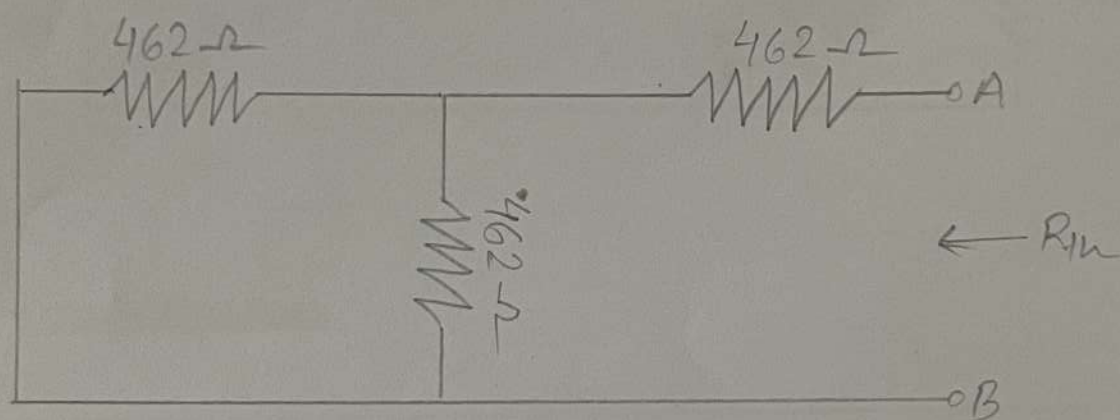


Figure: To find R_{th}

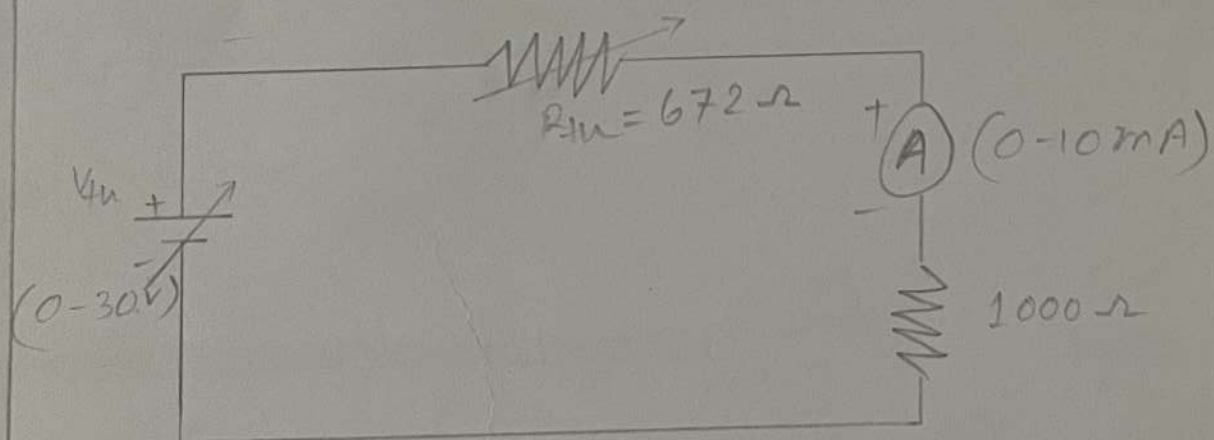
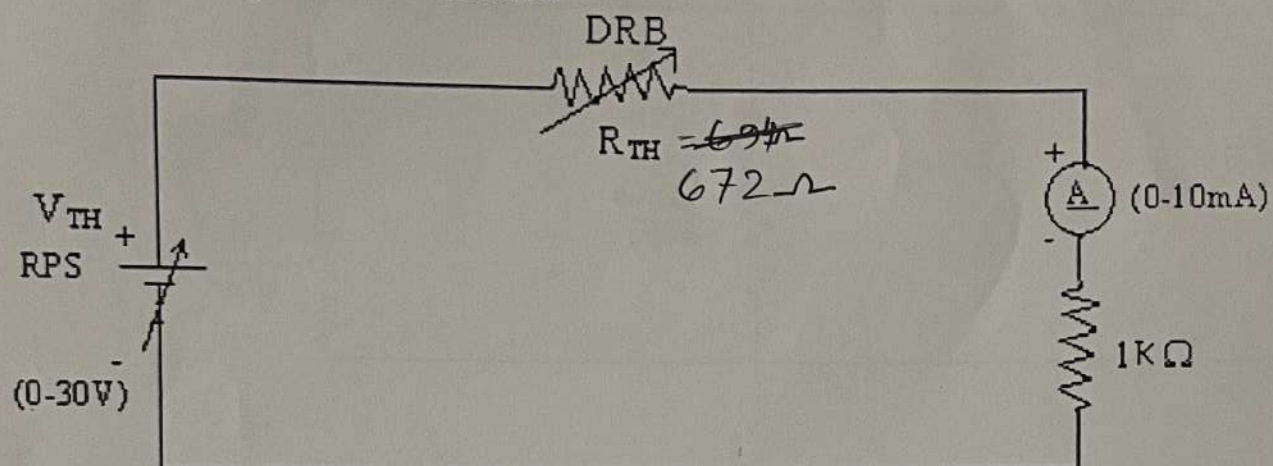


Figure: Thevenin's Equivalent Circuit

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Thevenin's Equivalent circuit:



Experimental Data:

Sl No.	Supply Voltage E (V)	V_{TH} (V)	R_{TH} (Ω)	I_L	
				Circuit 1 (mA)	Equivalent Circuit (mA)
Theoretical	10V	5V	693 Ω	2.95 mA	2.95 mA
Practical	10V	4.92 V	694 Ω	3 mA	3.8 mA 2.9 mA

Calculation: (For theoretical values V_{TH} and R_{TH})

Discussion:

Calculation:

$$R_{th} = \left(\frac{1}{462} + \frac{1}{462} \right)^{-1} + 462 = 693 \Omega$$

$$V_{th} = V \times \frac{R_2}{R_1 + R_2} = 10 \times \frac{462}{462 + 462}$$
$$= 5 \text{ V}$$

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{5}{693 + 1000}$$
$$= 2.95 \text{ mA}$$

Discussion:

The experiment verified the principle of Thevenin's Theorem. The load currents were nearly identical, confirming that equivalent circuit accurately modeled the original. However, a major quantitative error exists, as the theoretical current is highly inconsistent with the measured I_L values. This suggests significant errors in the practical current measurements relative to the reported V_{th} and R_{th} .